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mathematical methods for Data Mining

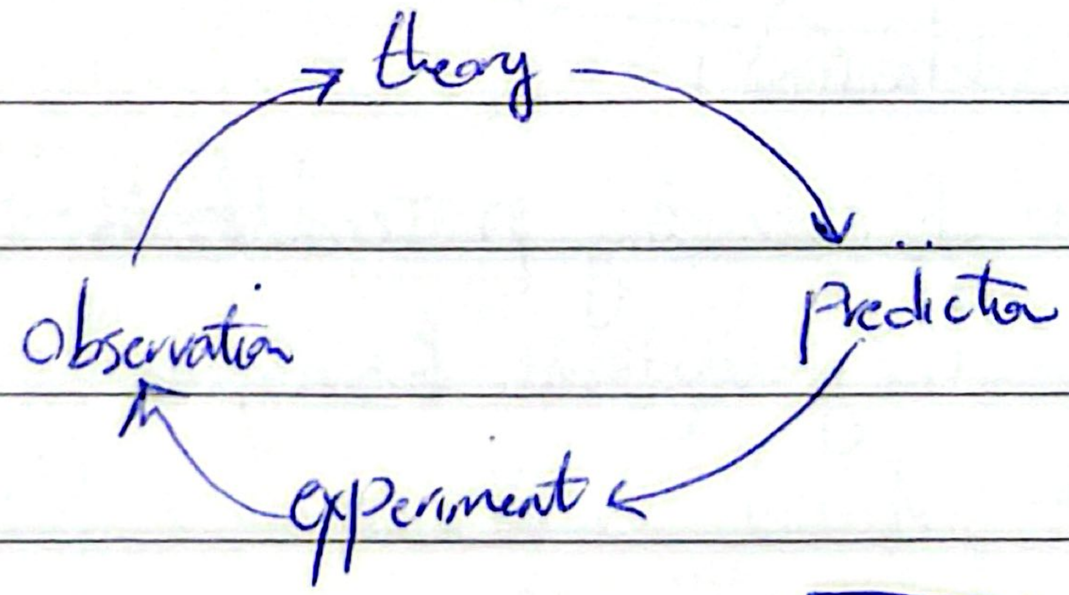
- [ ] 1. Key Ideas from Lecture 1
- [ ] 2. P1 Birthday problem
- [ ] 3. P2 IDF
- [ ] 4. P3 TF-IDF
- [ ] 5. Mistakes / things to redo

① key ideas from lecture 1 :

A. The Scientific Method :

- Uses objective experimentation to { Predict / Verify / refute } an assertion made by a theory
- Based on the results of the experiments, the theory is modified

(example : email spam detection)



* example of when we don't have good theory :

we can't easily write an exact mathematical rule that says wether a picture contains a cat.

So ⇒ Observed examples/data → learn approx. rule/function → Predict New Cases !

* Any computer programme is a function (rule)

⇒ So the question is:

how can we learn the hidden rule that connects inputs to outputs ?

⇒ Example :

- Input: { house size, location, number of rooms } → Output : Predicted price
- ...

(The model learns an approximate function)

features ? → Price

B. Data Mining (definition) :

* Process of extracting patterns from large datasets

by combining methods from { statistics, artificial intelligence, database management }

- Results of data mining { models, patterns }

example : Supermarket → those who bought chees and milk → will buy bread

- Machine Learning:

The ability to teach a computer without explicitly programming it.

(*Some regard data mining and machine learning as synonyms)

example: { movie recommendations are hard to hand-code bc ? taste? is messy
resume detection ; easier with simple keyword rules

{ machine learning helps when:
we do not know what we are looking for in data.
for example → it is unclear what about the movies make certain people like or dislike a movie

⇒ On The Other Hand!
machine learning has not proved successful in situations where we can describe the goals of mining more directly → Resume

C. Five Main Types of Applications

1) Classification → Predict a category/class label (Spam Emails)
(label is categorical)
[gives a category]

2) Numerical prediction/regression → Predict a number (house feat ↓ Sales price)
(label is numerical)
[gives a number]

3) Association → Find items/events that occur together (cheese milk = bread)
(IF A THEN B)

~~Unsupervised techniques~~

4) Clustering → Group similar objects when there is no label (group customers by behavior)
(Unsupervised learning)
(Unsupervised → label is unknown)

5) Ranking → Put items in an ordered list by importance/relevance (Google ranks Search results)
(Often linked to web/search/pageRank)
[gives an ordered list]

* Supervised Vs Unsupervised:

- features + known labels → learn prediction
- features only → discover structure/groups

* Statics vs Data Mining :

1) Data Mining uses data usually collected for some other purpose
2) Data mining process may not always be connected to data acquisition process.
3) In statistics data is often collected to answer specific questions

↓ all somehow the same!

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* Large Datasets :

1_ if datasets were small → we would use "statistical exploratory analysis"

2_ { large to statistician → has hundred to a thousand
large to data miner → millions or billions of data points

~~Many statistical methods require some type~~

3_ Problems with exceedingly large datasets → beyond traditional statistics problems
(disk access, efficiency, ...)

* 4. Many statistical methods require exhaustive search

As datasets becom larger → search becomes computationally more prohibitive

two ways { ① more records
② more variables

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~~Attacking every non-empty~~

* For p variables, checking every non-empty subset requires $2^p - 1$ searches.
(looking at all subsets of variables = $2^p - 1$)

5- Curse of dimensionality: exponential rate of growth in the number of unit cells in the data space
~~(unit cell: a cell with different features)~~

* 6- when a set of variables (features) is large ($X_1, X_2, \ldots, X_p$) ← large
then → high probability of finding a X that correlates with Y even if no correlation exists in the domain

→ Fake Correlations

(how it is related to data mining vs statistics;

- statistics:

start with a question → design data collection
↳ collect relevant data → test hypothesis

- Data mining

Huge existing dataset → search for interesting patterns
↳ build models/rules

⇒ Big Data gives more opportunities to discover useful pattern.
BUT ALSO more opportunities to discover nonesense patterns.

D. Bonferroni

* Statistical limits on Data Mining:

Bonferroni's principle (basically a warning about false discoveries in huge data)

⇒ it says:

if you search through a massive amount of data looking for rare/unusual events, some rare-looking events will appear just by chance, even if nothing meaningful is happening.

* informal rule:

Calculate the expected number of occurances under randomness, if this expected number is much larger than the number of real cases you hope to find most findings are probably bogus (nonesense)

~~→ Don't Bother gathering data~~

* bogus: a statistical artifact rather than evidence of what you are looking for.

Total Information Awareness example:

(?) if you search through enough innocent people's data, won't many innocent people accidentally look sus?!

→ Ans: yes! Unless the suspecious pattern is defined extremely narrowly and carefully

⇒ Bonferroni idea:

if there were no real Signal, how many matches would random chance still produce?

⇒ formula: $E = n \times p$

- E = expected number of random/bogus occurances
- n = number of opportunities/searches/trials
- p = probability that one opportunity matches by chance

(even if everyone is random/innocent, we expect about E people to match the pattern by chance)

* in the Bush example,

E = huge number of opportunities × Small probability

⇒ Still huge!

↓

Significantly larger than real instances

when the event is rare and not too broad, E becomes small

The "Bonferroni test":

1) Step 1: Define the event you are searching for
2) Step 2: Count the number of opportunities
3) Step 3: Find the probability that one opportunity matches by chance

Birthday paradox:

$$E(M_n) = \binom{k}{n} \times \frac{1}{N^{n-1}}$$

(k objects in N boxes, expected number of n-tuples in the same box)

(that's just the p ← هنا كتبدي n بكس ← اصل الـ k كتبدي)

*why it's called a paradox: you don't expect so many pairs from a class of 23 have shared birthdates

but $\binom{23}{2}$ is already a big number: 253

253 chances for a birthday match

example:

* Student 1, Student 2, Student 3, ...

$\Rightarrow$ random variable: $X_{1,2,3}$

where:

$$X_{1,2,3} = \begin{cases} 1, & \text{if student 1,2,3 have same birthday} \\ 0, & \text{otherwise} \end{cases}$$

* Now, for every possible triple:

$$X_{1,2,3}, X_{1,2,4}, X_{1,2,5}, \dots$$

Then the total number of matching triple:

$$M_3 = \sum_{\text{all triples}} X_{i,j,k} \qquad \left(X_{i,j,k} \text{ is } \begin{matrix} 0 \\ 1 \end{matrix}\right)$$

$\Rightarrow$

$$M_m = \sum_{\text{all } m\text{-tuples}} X_{i_1 \dots i_m}$$

$$X_{i_1 \dots i_m} = \begin{cases} 1 & \text{if these } m \text{ students shared birthday} \\ 0 & \text{otherwise} \end{cases}$$

if $K > (m-1)N \Rightarrow$ at least one box contains m objects (مبدأ الحمام)

=> Expectation of M_m:

$$E(M_m) = \binom{k}{m} \frac{1}{N^{m-1}}$$

$\binom{k}{m}$ Counts the opportunities

$\frac{1}{N^{m-1}}$ chance of chosen opportunity success.

$$\binom{k}{m} = \frac{k!}{m!\,(k-m)!} = \frac{k(k-1)(k-2)(k-3)\dots(k-m+1)}{m!}$$

Another example of Bonferroni idea:

Helmholtz principle:

Humans tend to notice a pattern when that pattern seems unlikely to have happened purely by chance

=> rare-looking pattern + many opportunities = maybe not meaningful

(There are two boxes with dots. The idea is that when there are too many dots eye might start seeing patterns, clusters, accidental shapes).

=> Bottom line ~~we~~ We Perceive Structure

⇒ O: Observed number of occurrences

E: Expected number of occurences under randomness

(1) $O \approx E$:

Weak evidence. Probably just random coincidence

(2) $O \gg E$:

Potentially Strong evidence

(randomness predicted E we saw way more → suspect non random)

(3) $E < 1$ and $O \geq 1$:

Strong evidence

E. labelled Vs unlabelled data

- Table → raws: instances, Samples, examples
  → clumns: features, variables, attributes

Tabular data → (1) labelled/(2) unlabelled data → supervised/unsupervised learning

i) For the labelled kind of data, there is a specially designated attribute and the aim is to use the data given to predict that attribute for instances that have not yet been seen → Supervised Learning

* designated attr. → Categorical (classification) → one of a number of distinct vals
  → numerical (regression)

G.

2) Data that is unlabelled desn't have any designated attribute

↳ Unsupervised Learning:

The aim is to extract the most information we can from the data available

(Association and clustering)

- Outliers / Anomalies:

An object that does not comply with the general behaviour or model of the data

("Napoleon Dinimite" in the netflix competition)

- many data mining methods discard outliers

↳ because they wanna learn general pattern

+ Sometimes rare case is not useless noise.

- Example: Outlier analysis may uncover fraudulent usage of credit cards (by unusually large purchases or location frequency)

↓

Anomaly mining = Outlier Analysis

⇒

Lecture 1: Data mining is about: "extracting patterns from large datasets"

↳ { text mining
web mining

08.05.2026

TF-IDF (How important is this word for this particular doc compared with the whole collection)

IDF Computation

(Text mining / Document representation - side of data mining)

- Document-term matrix:

Document	cat	dog	cats	fish	meat
D_1	1	0	1	1	0
D_2	0	1	1	0	1
D_3	1	0	0	0	0

- IDF: Inverse Document Frequency:

a statistical measure that evaluates how rare or common a word is, across a collection of documents.

↳ How rare is word i across the whole document collection?

$$IDF_i = \log_2 \left(\frac{N}{n_i} \right)$$